

Cyril Creque-Sarbinowski

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■ External Resource

■ Slides/Notes

■ Paper

Education

Johns Hopkins University (JHU)

Ph.D in Physics & Astronomy

Advisor: Marc Kamionkowski

Baltimore, MD

2017-2022

Massachusetts Institute of Technology (MIT)

B.S. in Physics & Mathematics

Cambridge, MA

2013-2017

Positions Held

Center for Computational Astrophysics (CCA)

Flatiron Research Fellow

New York, NY

August 2022-Present

New York University (NYU)

Visiting Scholar

New York, NY

September 2022- September 2024

Awards, Scholarships, Fellowships

National Science Foundation

Graduate Research Fellow

2019-2022

Gates Millenium Scholars Program

Scholar

2013-2022

JHU Kelly Miller Fellowship

Fellow

2017-2019

Teaching/Outreach

Sound Plus Science

Chief Facilitator & Mentor, Developed Curriculum [Sample Exercise]

Civic Hall

March 2024-Present

GEM Annual Conference

Flatiron Institute Booth Staff

September 2023, 2024

ASTRO 10000: Basic Concepts in Astronomy

Guest Lecturer

Hunter College

May 2023

PHYS 111: Introductory Physics

Virtual Guest Lecturer

Moravian University

Spring 2022

PHYS2300: Quantum Theory of Fields I

Guest Lecturer

Brown University

Spring 2022

AS.171.107 General Physics for Physical Science Majors

Teaching Assistant

JHU

Fall 2017

Lecturer: Prof. Robert Leheny & Prof. Rosemary Wyse

AS.173.111 General Physics Laboratory I

Section Head

JHU

Fall 2017

Selected Conferences, Workshops, Seminars

* = *Invited*, † = *Organizer*

2024.....		
Workshop	Cosmology in the Adriatic – From PT to AI, BCCP, Ruder Bošković Institute, Faculty of Science in Split	July
Seminar*	Cosmology Group Meeting, Perimeter Institute	July
Program	Dark Matter Theory, Simulation, & Analysis in the Era of Large Surveys, Kavli Institute for Theoretical Physics (KITP), Affiliate	May-June
Workshop	The Future of Dark Matter, Columbia University/Flatiron Institute	April
Symposium†	Caribbean Future of Science Symposium	March
Seminar*	KICP Seminar, University of Chicago	January
2023.....		
Workshop*	Large-Scale Parity Violation Workshop, Academia Sinica Institute of Astronomy & Astrophysics (ASIAA)	December
Workshop†	Parity Violation from Home	October
Talk*	Cosmology Talks	August
Workshop*	Ending Inflation & the Hot Big Bang, Stonybrook University	June
Workshop*	Modified Gravity Program, Illinois Center for Advanced Studies of the Universe	May
Symposium*	Future Faculty Symposium in Physics, Princeton University	April
Seminar*	Quantum Gravity Seminar, Perimeter Institute	April
Seminar*	Cosmology Seminar, Max Planck Institute for Astrophysics	April
Seminar*	Cosmology Seminar, Tufts/MIT	February
Seminar*	IDEA Seminar, The Brown Theoretical Physics Center (BTTC)	February
2022.....		
Conference	NSBP Annual Conference	November
Workshop*	Guest Expert, Cosmology Talks Parity-Violation Mini-Workshop	November
Seminar*	Theory Group Seminar, Imperial College London	November
Talk*	AstroParticle Lunch, The Ohio State University	September
Talk	Early Universe/Cosmology Lunch Discussion, Princeton University/Institute of Advanced Study	September
Workshop†	NeXus Workshop, CCA, Flatiron Institute	August
Workshop	Large-Scale Structure Cosmology beyond 2-Point Statistics, Aspen Center for Physics	May-June
Seminar*	Brown University	March
$t \leq 2021$		
Workshop	NeXus Workshop, CCA, Flatiron Institute	June 2021
Conference	Phenomenology 2021	May 2021
Workshop	XIX International Workshop on Neutrino Telescopes	February 2021
Conference	Neutrino 2020	June 2020
Seminar*	Physics Seminar, Bowdoin College	November 2019
Conference	NSBP Annual Conference	November 2019
Conference	Cosmic Controversies, Kavli Institute for Cosmological Physics (KICP)	October 2019
Symposium	Symposium in Honor of the Legacy of Vera Rubin, Georgetown University	June 2019
Workshop	Novel Ideas for Dark Matter, Princeton Center for Theoretical Science (PCTS)	January 2019
Conference	NSBP Annual Conference	November 2018

Conference	Identification of Dark Matter, Brown University	July 2018
Workshop	Second Annual Line-Intensity Mapping Workshop (IM@Hopkins), Johns Hopkins University (JHU)	July 2017
Conference	Neutrinos & Light Particles in Cosmology, Berkeley Center for Cosmological Physics (BCCP)	June 2016
Conference	NSBP Annual Conference	February 2015

Professional Activities

Referee	Phys. Rev. D (2023–), Phys. Rev. Lett. (2023–) MNRAS (2022–), Philos. Trans. R. Soc. A (2024–)	
Facilitator	NeXus, Lead Organizer and Mentor	2022–Present
Organizer	Particles Vs. New Probes (P Vs. NP), Lead Organizer Caribbean Future of Science Symposium, SOC Parity Violation from Home, Co-Organizer NeXus Workshop, Lead Organizer	March 2025 March 2024, 2025 Oct. 2023, Nov. 2024 August 2022
Review	GEM Fellowship Program	November 2023, 2024
Panelist		

Publications [13 ref. journal pub. (10 1st-author) +1 submitted, 300+ citations, h-index: 8]

- [1] S. Alexander, H. Bernardo, and **C. Creque-Sarbinowski**, “Nontriviality of dynamical Chern-Simons gravity and the standard model,” *Phys. Rev. D* **110** no. 2, (2024) 025015, [arXiv:2403.15657 \[hep-th\]](#).
- [2] S. Alexander, **C. Creque-Sarbinowski**, H. Gilmer, and K. Freese, “Higgs Inflation and the Electroweak Gauge Sector,” [arXiv:2306.04671 \[hep-ph\]](#).
- [3] J. Flitter, **C. Creque-Sarbinowski**, M. Kamionkowski, and L. Dai, “Magnetic fields from compensated isocurvature perturbations,” *Phys. Rev. D* **107** no. 10, (2023) 103536, [arXiv:2304.03299 \[astro-ph.CO\]](#).
- [4] **C. Creque-Sarbinowski**, S. Alexander, M. Kamionkowski, and O. Philcox, “Parity-violating trispectrum from Chern-Simons gravity,” *JCAP (20th Anniversary Special Issue)* **11** (2023) 029, [arXiv:2303.04815 \[astro-ph.CO\]](#).
- [5] B. Zhou, L. Reali, E. Berti, M. Çalışkan, **C. Creque-Sarbinowski**, M. Kamionkowski, and B. S. Sathyaprakash, “Subtracting compact binary foregrounds to search for subdominant gravitational-wave backgrounds in next-generation ground-based observatories,” *Phys. Rev. D* **108** no. 6, (2023) 064040, [arXiv:2209.01310 \[gr-qc\]](#).
- [6] B. Zhou, L. Reali, E. Berti, M. Çalışkan, **C. Creque-Sarbinowski**, M. Kamionkowski, and B. S. Sathyaprakash, “Compact Binary Foreground Subtraction in Next-Generation Ground-Based Observatories,” [arXiv:2209.01221 \[gr-qc\]](#).
- [7] S. Alexander and **C. Creque-Sarbinowski**, “UV completions of Chern-Simons gravity,” *Phys. Rev. D* **108** no. 10, (2023) 104046, [arXiv:2207.05094 \[hep-ph\]](#).
- [8] **C. Creque-Sarbinowski**, J. Hyde, and M. Kamionkowski, “High-energy astrophysical neutrinos from cosmic strings,” *Phys. Rev. D* **107** no. 12, (2023) 123019, [arXiv:2206.06377 \[hep-ph\]](#).
- [9] **C. Creque-Sarbinowski**, M. Kamionkowski, and B. Zhou, “Seeking neutrino emission from AGN through temporal and spatial cross-correlation,” *Phys. Rev. D* **105** no. 12, (2022) 123035, [arXiv:2111.08012 \[astro-ph.HE\]](#).

- [10] **C. Creque-Sarbinowski**, M. Kamionkowski, and B. Zhou, “AGN variability in the age of VRO,” *Astrophys. J.* **941** (2022) 41, [arXiv:2110.13149 \[astro-ph.GA\]](#).
- [11] **C. Creque-Sarbinowski**, J. Hyde, and M. Kamionkowski, “Resonant neutrino self-interactions,” *Phys. Rev. D* **103** no. 2, (2021) 023527, [arXiv:2005.05332 \[hep-ph\]](#).
- [12] **C. Creque-Sarbinowski**, L. Ji, E. D. Kovetz, and M. Kamionkowski, “Direct millicharged dark matter cannot explain the EDGES signal,” *Phys. Rev. D* **100** no. 2, (2019) 023528, [arXiv:1903.09154 \[astro-ph.CO\]](#).
- [13] **C. Creque-Sarbinowski** and M. Kamionkowski, “Searching for Decaying and Annihilating Dark Matter with Line Intensity Mapping,” *Phys. Rev. D* **98** no. 6, (2018) 063524, [arXiv:1806.11119 \[astro-ph.CO\]](#).
- [14] **C. Creque-Sarbinowski**, S. Bird, and M. Kamionkowski, “Cross-correlation between thermal Sunyaev-Zeldovich effect and the integrated Sachs-Wolfe effect,” *Phys. Rev. D* **94** no. 6, (2016) 063519, [arXiv:1606.00839 \[astro-ph.CO\]](#).

References

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